

CS4084

Quantum Computing

Week 16 – Lecture 31-32

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Outline

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- 2 The Need for Error Correction: Classical vs. Quantum
- 3 Classical repetition codes
- 4 Repetition codes for qubits
- 5 The 9-Bit Shor Code: Complete Protection
- 6 What are Channels? The Big Picture
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Course Retrospective: From Foundations to Systems

The Building Blocks of Computation:

- **Classical vs. Quantum Domains:** Transitioning from bits to qubits; understanding the shift from deterministic states to **Superposition** and **Entanglement**.
- **System Complexity:**
 - **Single Systems:** State vectors, Bloch Sphere representation, and measurement postulates.
 - **Compound Systems:** Tensor products, the No-Cloning Theorem, and Bell States.
- **The Language of Unitaries:** Mapping evolution through **Unitary Matrices** ($U^\dagger U = I$) and the physical implementation of **Quantum Gates** (H, X, CNOT, T).
- **Computational Projects:** Practical application of theoretical concepts through simulated quantum circuits and algorithmic design.

The Path to Fault-Tolerance I

Quantum Advantage & Complexity:

- **The Query Model:** Establishing efficiency by counting Oracle calls (Deutsch-Jozsa, Simon's, and Grover's).
- **The Power of Pattern Matching:**
 - Using **Phase Kickback** and **QFT** to solve hidden subgroup problems.
 - **Shor's Algorithm:** The practical application of Period-Finding to factorize integers.

The Path to Fault-Tolerance II

Final Week: Correcting Quantum Errors

- **The NISQ Reality:** Understanding how **Depolarizing Noise** and **Phase Damping** threaten the validity of long-circuit computations.
- **Quantum Error Correction (QEC):**
 - **The Challenge:** How to measure errors without collapsing the wavefunction (The Zeno Effect vs. Syndrome Measurement).
 - **Redundancy:** Encoding one logical qubit into multiple physical qubits (e.g., The 3-Qubit Bit-Flip Code).
 - **The Goal:** Achieving **Fault-Tolerance**—where the rate of error correction exceeds the rate of decoherence.

The Need for Error Correction: Classical vs. Quantum

1. Classical Resilience: "Macroscopic Stability"

- **High Signal-to-Noise Ratio:** Classical bits are realized by billions of electrons; a few stray charges won't flip a bit.
- **Error Type:** Limited to the discrete **Bit-Flip** ($0 \leftrightarrow 1$).
- **Solution:** Simple redundancy (e.g., Repetition Code: $0 \rightarrow 000$).

2. Quantum Fragility: "The Microscopic Challenge"

- **High Sensitivity:** Qubits are individual particles (atoms, photons, ions). The slightest environmental interaction causes **Decoherence**.
- **Continuous Errors:** Unlike binary bits, quantum states can suffer from **Phase Flips** ($|+\rangle \leftrightarrow |-\rangle$) and small rotation errors (e.g., $e^{i\epsilon Z}$).
- **Three Fundamental Barriers to QEC:**
 - 1 **No-Cloning Theorem:** We cannot copy $|\psi\rangle$ to create redundancy.
 - 2 **Measurement Collapse:** Measuring a qubit to find an error destroys the superposition.
 - 3 **Error Diversity:** We must correct both Bit-Flips (X) and Phase-Flips (Z).

Conclusion: QEC requires "Syndrome Measurements" that reveal the error without revealing the data.

Foundation: The Classical Repetition Code

The Concept: "Majority Voting" To protect a single bit against a flip, we map the logical bit to a higher-dimensional physical space using redundancy.

- **Encoding:** $0 \rightarrow [000]$, $1 \rightarrow [111]$
- **The Noise Model:** Assume each bit has a small probability p of flipping.
- **Decoding (Recovery):** We take a **Majority Vote** of the three bits.

Example Scenario:

Sent	Received (Error)	Majority Vote	Result
[000]	[010] (Bit 2 flipped)	0	Corrected!
[111]	[110] (Bit 3 flipped)	1	Corrected!
[000]	[110] (Two bits flipped)	1	Failure

Transition to Quantum (The 9-Bit Code):

- **Classical:** Only protects against X (Bit-flips).
- **Quantum Problem:** We must handle **Bit-flips**, **Phase-flips** (Z), and **Both** (Y).
- **The Shor Logic:** We nest this repetition: use 3 qubits for bit-flips, then repeat that entire block 3 times to handle phase-flips.

Classical repetition codes

Repetition codes are very basic examples of error correcting codes. The idea is simply to repeat each bit multiple times.

3-bit repetition code

Encoding

$0 \mapsto 000$

$1 \mapsto 111$

Decoding

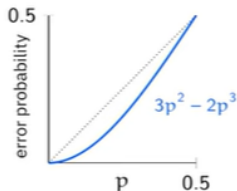
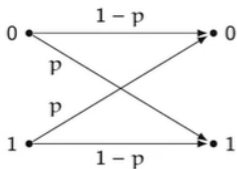
$abc \mapsto \text{majority}(a, b, c)$

This code corrects up to one **bit flip** on any of the three bits used for encoding.

Continue..

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Suppose each bit is sent through a **binary symmetric channel** that flips a bit with probability p .

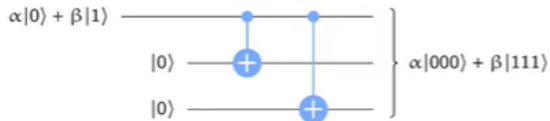


Repetition codes for qubits

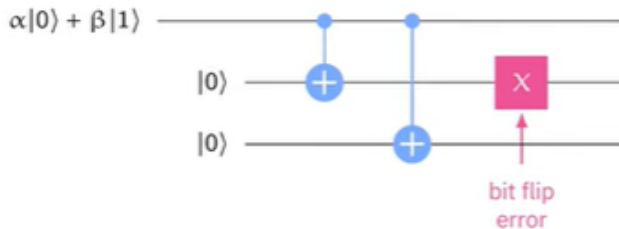
The 3-bit repetition code can be used to encode a qubit:

$$\alpha|0\rangle + \beta|1\rangle \mapsto \alpha|000\rangle + \beta|111\rangle$$

This circuit performs the encoding:



Lets suppose Error occurred after Encoding



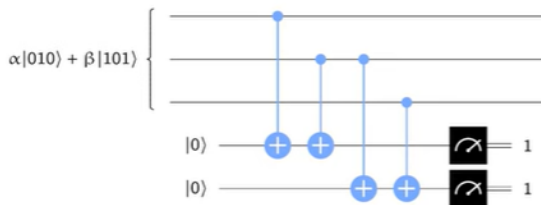
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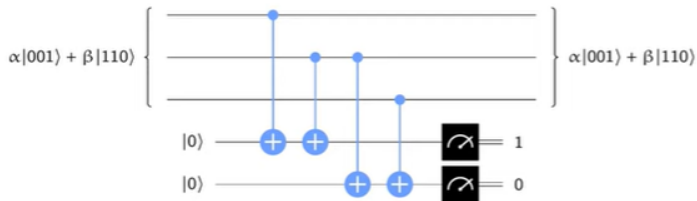
$$\alpha|0\rangle + \beta|1\rangle \mapsto \alpha|000\rangle + \beta|111\rangle$$



We can identify the location of a single bit-flip with this circuit:

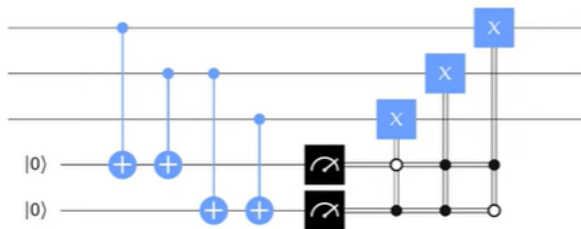


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State	Syndrome	Correction
$\alpha 000\rangle + \beta 111\rangle$	00	$1 \otimes 1 \otimes 1$
$\alpha 100\rangle + \beta 011\rangle$	10	$X \otimes 1 \otimes 1$
$\alpha 010\rangle + \beta 101\rangle$	11	$1 \otimes X \otimes 1$
$\alpha 001\rangle + \beta 110\rangle$	01	$1 \otimes 1 \otimes X$

Only 1 bit Flip error with possible correction



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Correcting Phase Flip Errors (Z)

The Challenge: A Phase Flip error $Z|\psi\rangle$ turns $\alpha|0\rangle + \beta|1\rangle$ into $\alpha|0\rangle - \beta|1\rangle$. In the standard basis, $|0\rangle$ and $|1\rangle$ remain unchanged, making the error "invisible" to standard bit-flip checks.

- **The Solution: Change of Basis** Using the identity $HZH = X$, we can treat a phase flip in the Z -basis as a bit flip in the X -basis $\{|+\rangle, |-\rangle\}$.
- **Phase Flip Encoding ($n = 3$):** Instead of $|000\rangle$, we encode the logical state using $|+\rangle$ states:

$$|0\rangle_L \rightarrow |+++\rangle, \quad |1\rangle_L \rightarrow |--\rangle$$

- **Error Detection:** A Z error on any physical qubit flips $|+\rangle \leftrightarrow |-\rangle$.
 - **Example:** $|+++\rangle \xrightarrow{Z_2} |+-+\rangle$
 - **Syndrome Measurement:** We check the parity of adjacent qubits in the X -basis (using X_1X_2 and X_2X_3 observables).

The Recovery Process:

- 1 Apply Hadamard gates to return to the computational basis.
- 2 Perform a standard "Majority Vote" check.
- 3 Apply X correction if needed, then Hadamard back.

The 9-Bit Shor Code: Complete Protection I

The Concept: Concatenation The Shor code "nests" the phase-flip code and the bit-flip code to handle any single-qubit error. It uses 9 physical qubits to represent 1 logical qubit.

- **Step 1: Phase-Flip Encoding (Outer Layer)** The logical state is first mapped to the X -basis $\{|+\rangle, |-\rangle\}$:

$$|\psi\rangle \rightarrow \alpha|+++ \rangle + \beta|--- \rangle$$

- **Step 2: Bit-Flip Encoding (Inner Layer)** Each $|+\rangle$ and $|-\rangle$ is further encoded using the 3-bit repetition code:
 - $|+\rangle \rightarrow \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$
 - $|-\rangle \rightarrow \frac{1}{\sqrt{2}}(|000\rangle - |111\rangle)$

The 9-Bit Shor Code: Complete Protection II

The Fully Encoded Logical States:

$$|0\rangle_L = \frac{1}{2\sqrt{2}}(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)$$

$$|1\rangle_L = \frac{1}{2\sqrt{2}}(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)$$

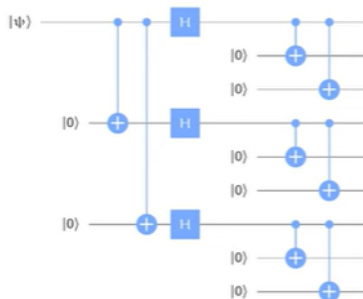
Capabilities:

- **Bit-Flip (X):** Corrected by the inner triplets (majority vote within each block of 3).
- **Phase-Flip (Z):** Corrected by the outer layer (comparing signs between the three blocks).
- **Combined ($Y = iXZ$):** Since it can handle X and Z independently, it naturally corrects Y errors.

Historical Note: This proved that quantum error correction is possible despite the No-Cloning theorem.

9 bit Shor code

This is the **concatenation** of the 3 qubit phase-flip and bit-flip repetition codes.



$$|0\rangle \mapsto \frac{1}{2\sqrt{2}}(|000\rangle + |111\rangle) \otimes (|000\rangle + |111\rangle) \otimes (|000\rangle + |111\rangle)$$

$$|1\rangle \mapsto \frac{1}{2\sqrt{2}}(|000\rangle - |111\rangle) \otimes (|000\rangle - |111\rangle) \otimes (|000\rangle - |111\rangle)$$

What are Channels? The Big Picture

In information theory, a **Channel** is any physical medium or process that carries information from a sender (Alice) to a receiver (Bob).

- **The Basic Function:** It maps an input state to an output state.
- **The Presence of Noise:** No physical channel is perfect. Interaction with the environment causes the signal to degrade, leading to errors.
- **Mathematical Role:** A channel is a mathematical transformation (a "map") that describes how information changes over time or distance.

Sender $\xrightarrow{\text{Transformation / Noise}}$ **Receiver**

In Quantum Computing, the "Channel" is often just the passage of time or the execution of a gate on real, noisy hardware.

Classical vs. Quantum Channels

Feature	Classical Channel	Quantum Channel
Unit of Info	Bits (0 or 1)	Qubits ($ 0\rangle$, $ 1\rangle$, or superposition)
Evolution	Stochastic (Probabilistic)	Completely Positive Trace Preserving (CPTP)
Noise Type	Bit Flip ($0 \leftrightarrow 1$)	Bit Flip, Phase Flip, and Decoherence
Observability	Can be monitored without change	Observation causes Wavefunction Collapse
Capacity	Max bits per use	Max qubits (or classical bits) per use

The Key Difference: Classical channels deal with **Probability Distributions**. Quantum channels deal with **Density Matrices** (ρ) and preserve the complex interference patterns between states.

Types of Channels: From Logic to Reality

Classical Channels:

- **Binary Symmetric Channel (BSC):** A bit flips with probability p .
- **Binary Erasure Channel (BEC):** A bit is lost (erased) with probability p .

Quantum Channels:

- **Unitary Channel:** Ideal, noise-free evolution ($\rho \rightarrow U\rho U^\dagger$).
- **Depolarizing Channel:** The qubit is "smashed" into white noise.
- **Dephasing Channel:** The qubit loses its "quantum phase" but keeps its $|0\rangle$ or $|1\rangle$ value.
- **Amplitude Damping:** Models energy loss (e.g., a photon being absorbed).

Modeling Quantum Channels: The Toolbox

We use three primary mathematical lenses to "see" a quantum channel:

① The Kraus Representation (Operator Sum):

$$\Phi(\rho) = \sum_k E_k \rho E_k^\dagger$$

The operators E_k describe the different things that "might happen" to the qubit due to noise.

② Stinespring Dilation (The Environment View):

Every noisy channel is just a "clean" Unitary interaction between our qubit and a huge Environment. We get the "noise" by simply ignoring the environment.

③ The Choi Matrix:

A way to turn a dynamic process into a static matrix. If we can measure the Choi Matrix, we know everything there is to know about the channel's errors.

Closing: This is Just the Beginning

While our semester ends here, the field of Quantum Information is vast and rapidly evolving. The foundations we have built—Unitaries, Gates, and Error Correction—are the entry points to the next frontier:

- **Quantum Machine Learning (QML):** Leveraging quantum speedup for pattern recognition, neural networks, and optimization.
- **Variational Algorithms (VQE/QAOA):** Using noisy devices (NISQ) to solve chemistry and logistical problems.
- **Quantum Networking:** Moving beyond single systems to a global Quantum Internet.

The journey does not end here; it scales up!

Thank you for a great semester!

Good luck!